From Zero to Task-Master

Hands-On Fine-Tuning of SLMs for Specialized Applications

A Cybersecurity Case-study

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Dream

MLPrague 2026

DREAM

Workshop Agenda & Outcomes

From theoretical foundations to local intelligence

Foundations

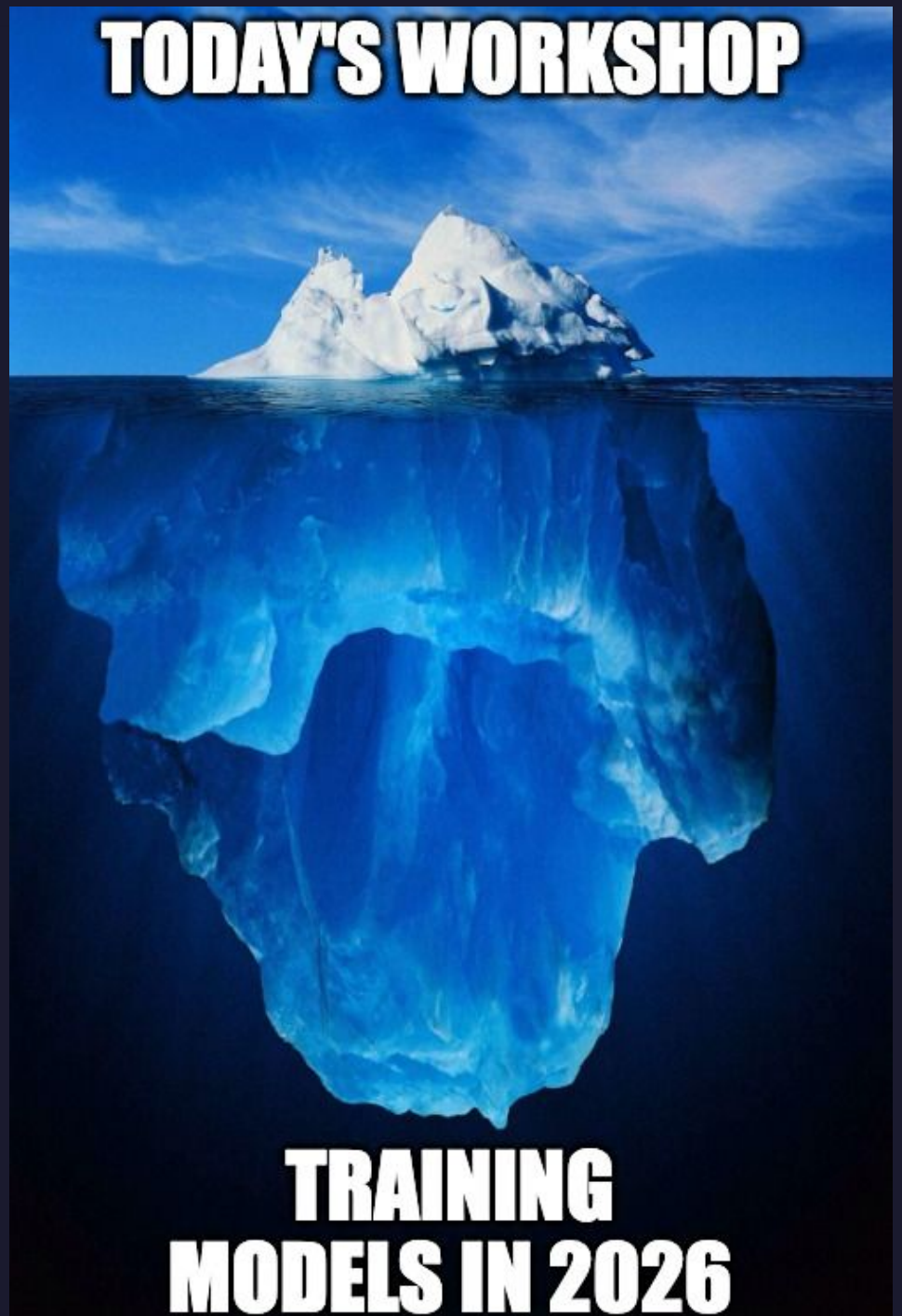
Deep dive into SLMs, local modes, fine-tuning fundamentals, and AI+CyberSec.

Hands-On Lab

Interactive session where we'll train our own model for a CyberSec task.

Key Takeaway / Outcome

Understand fine-tuning principles and master the ability to efficiently train your own specialized SLM


TODAY'S WORKSHOP
TRAINING
MODELS IN 2026

Small Language Models (SLMs)

High efficiency, maximum privacy, specialized performance.

Performance

Fast, efficient, strong - yet focused.

- Low latency
- Reduced compute
- Memory efficient

Privacy

Full local control for data privacy.

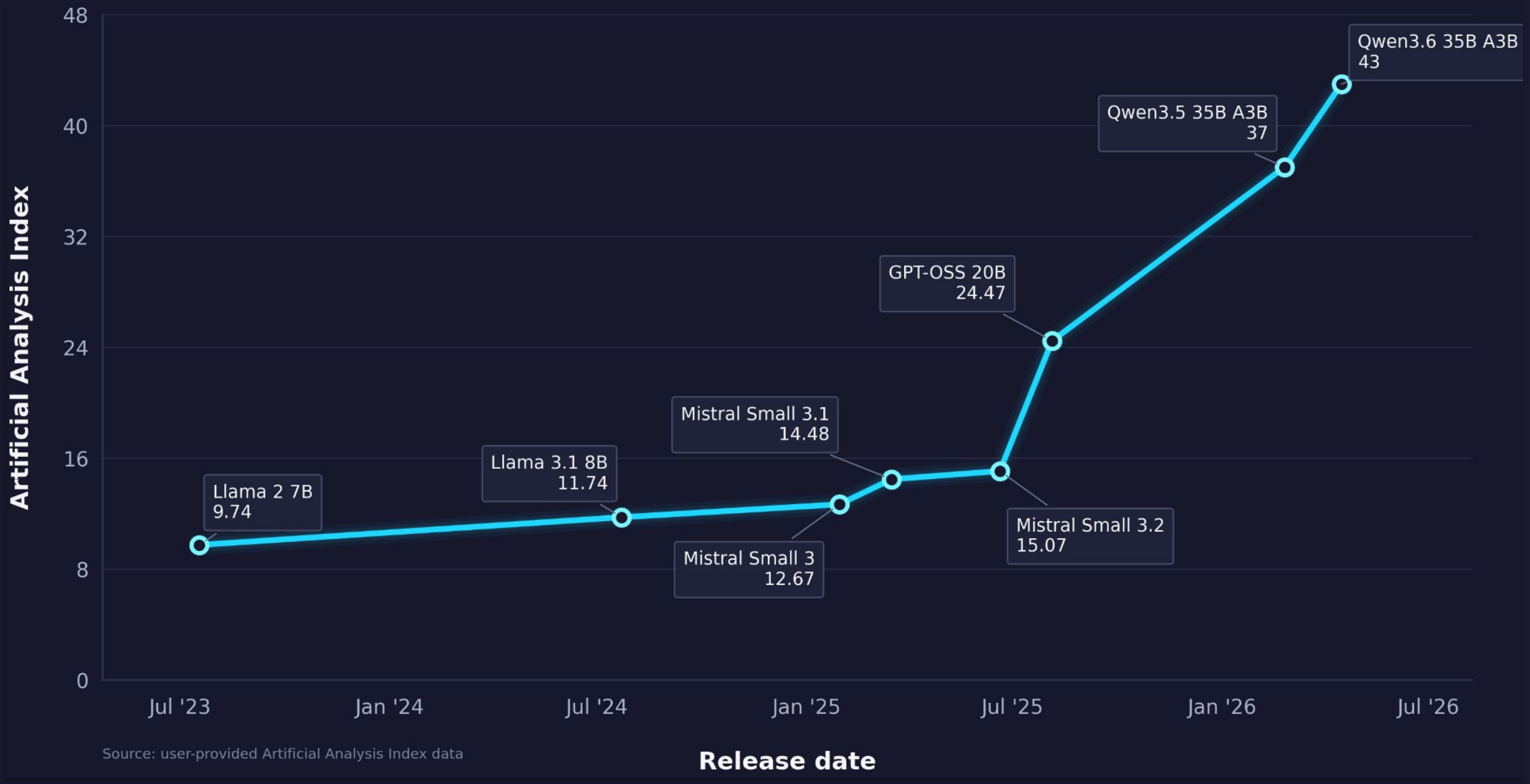
- Data sovereignty
- No cloud dependency
- Secure processing

Specialization

Excel at niche tasks with fine-tuning.

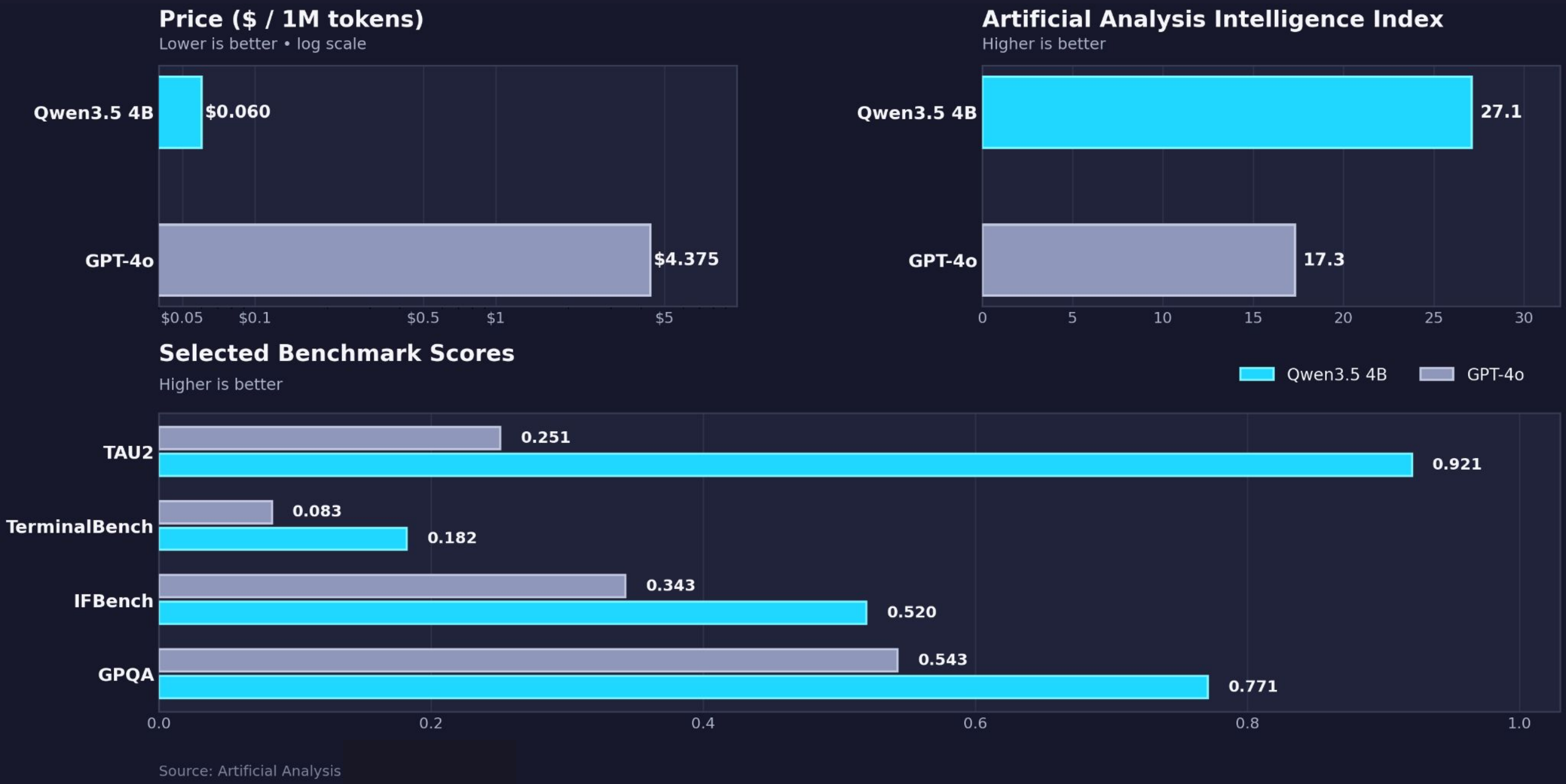
- Domain expertise
- Task specific accuracy
- Customization

SLMs Have Significantly Improved



Source: user-provided Artificial Analysis Index data

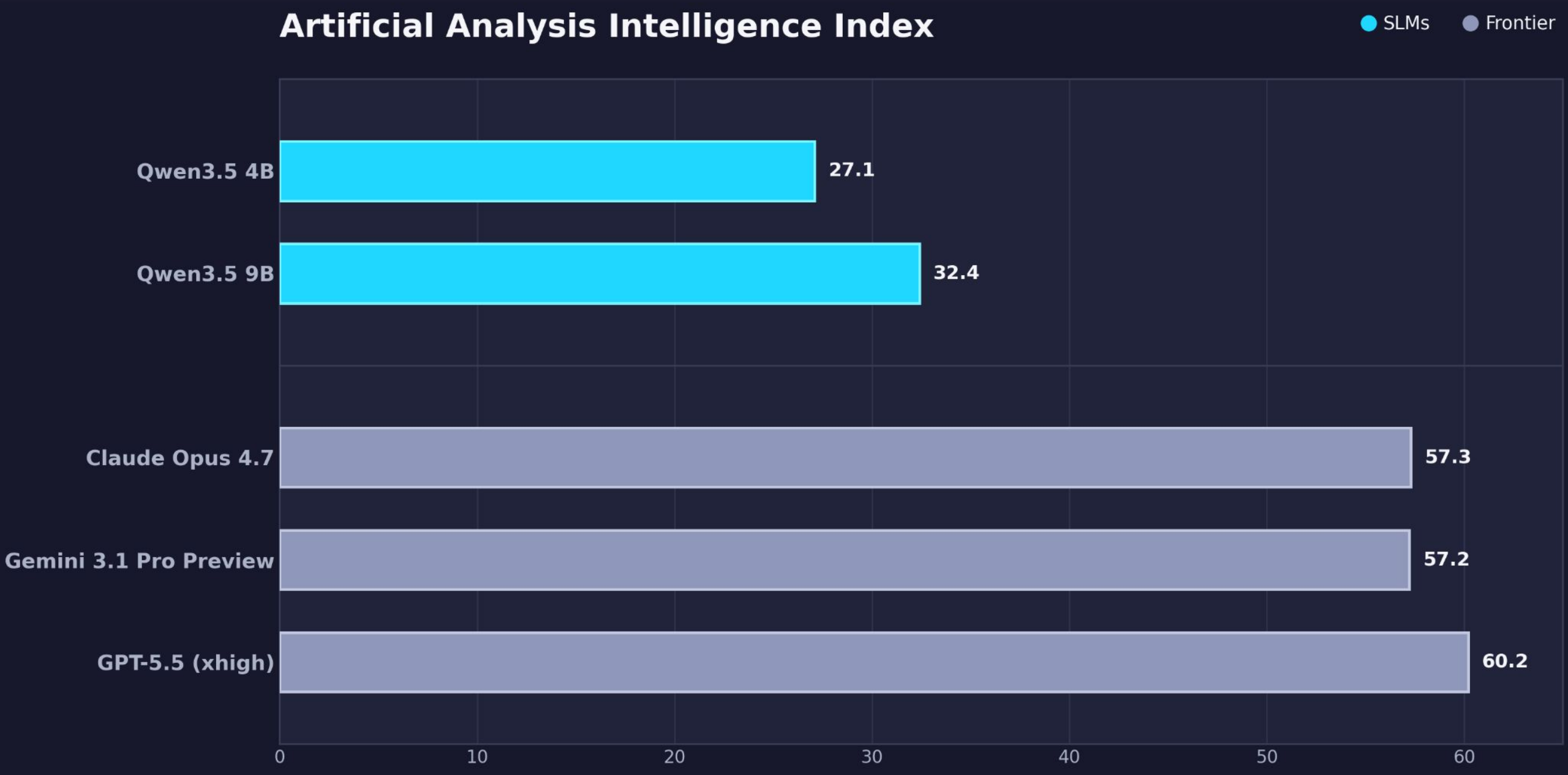
Yesterday’s Frontier - Today’s SLM



What Caused this Improvement?

- **Reasoning:** Models that were trained using RL use inference time-compute (thinking) to get better answers.
- **Efficiency:** Better hardware, optimizations, and mixture-of-experts architecture enable running larger models more efficiently.
- **Quality:** Better data, agentic training, synthetic data, and deep industry experience.

Today’s Frontier Is Still Far



Fine-tuning

Adapting a pre-trained model to specialize it for a specific task.

Pre-trained Model

Start from a foundation base model.

Data Collection

Collect training data for a specific task or domain.

Model Training

Train the model on the collected data.

LLM Internals

INPUT SENTENCE

Prague	is	the	capital	of	the	→	?
6836	318	262	3139	286	262		next

TOP NEXT-TOKEN PREDICTIONS

0	19.35%	16639	Czech
1	4.76%	15708	Kingdom
2	3.71%	12671	European
3	2.47%	1966	Country
4	1.13%	47371	Region
5	0.73%	9012	State

Pre-training

Massive Scale Training
Training on a massive scale - trillions of tokens of raw text.

Next-Token Prediction
The fundamental objective of the pre-training phase.

Foundation Base Model
Resulting model possesses vast knowledge and semantic understanding.

Capability Gaps
Base models lack specific refinement and conversational chat capabilities.

Supervised Fine-tuning (SFT)

What is SFT?

Supervised Fine-tuning is the process of training a pre-trained model on a labeled dataset of (instruction, response) pairs.

- **Goal:** Teach the model to follow specific formats or styles.
- **Mechanism:** Gradient descent updates weights to minimize difference between model output and ground truth labels.
- **Dataset:** High-quality, curated demonstrations of the desired behavior.

SFT in Action: Learning from Examples

User:
What is the capital of the Czech Republic?

Assistant
The capital of the Czech Republic is Prague.

User:
What river runs through the city?

Assistant
The Vltava River runs through Prague.

When to Fine-tune?

AI in Cybersecurity - Making Headlines

What is Claude Mythos and what risks does it pose?

New AI model "Claude Mythos" emerges as a significant focal point in cybersecurity.

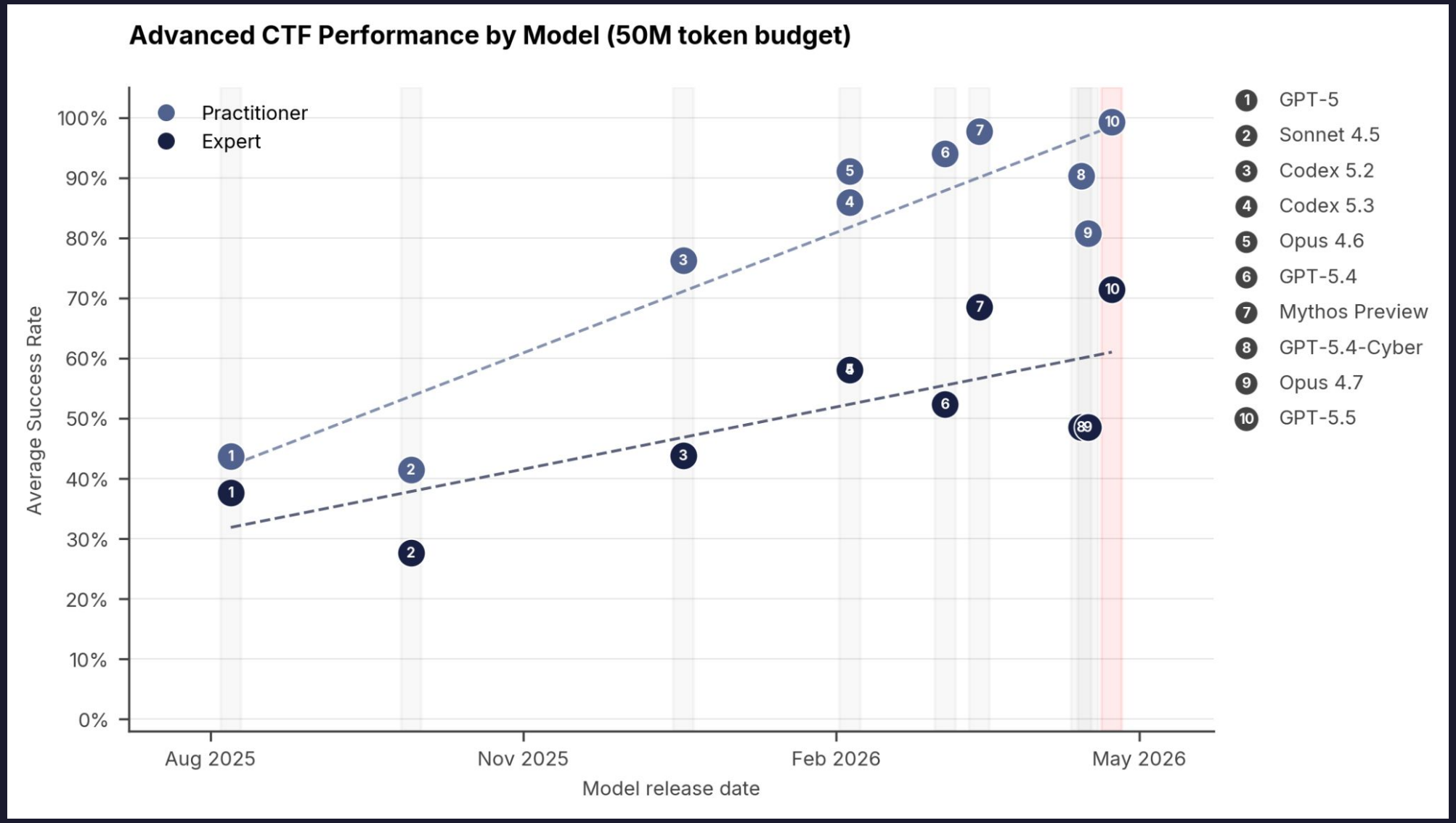
OpenAI's GPT-5.5 Matches Claude Mythos in Cyberattack Capabilities

AI Security Institute.

White House Questions Tech Industry on Defensive AI Use

Focus on cybersecurity resilience and the role of tech giants in defensive strategy.

AI in Cybersecurity



[AISI Cybersecurity Benchmark]

AI in Cybersecurity

A high-stakes, adversarial, data-rich domain

Fast Adversaries

Attackers iterate fast, faster than humans can write rules

Static Limits

Rule-based defenses miss novel and complex threats.

Complex Data

CyberSec data is messy and unstructured: emails, URLs, HTML, logs, code

Pattern Mastery

LLMs brings pattern recognition at massive scale

LLM Reasoning

Language models can read, verify, and reason over textual/code data.

Where AI Helps in Cybersecurity

Phishing Detection

Classify emails & web pages (today's focus)

Malware Classification

Identify malicious binaries and scripts

Anomaly Detection

Spot unusual network or login behavior

Red Teaming

Cloud, web and infrastructure Penetration Testing,

Vulnerability Discovery

Find security bugs in source code

Incident Response

Triage alerts and draft response playbooks

Each task benefits from a small, specialized, fine-tuned model

Hands-on Lab - Agenda

1. **The Problem & Dataset**
   Phishing detection - why it matters

2. **Baseline Evaluation**
   How does an out-of-the-box SLM perform?

3. **Exploration**
   Explore data, prompt engineering

4. **SFT Training**
   Fine-tune an SLM

5. **Evaluation & Wrap-Up**
   Compare before vs. after

1 The Problem: Phishing Detection

- ⚠️ Phishing attacks trick users into revealing sensitive information
- 🌐 Attackers clone legitimate websites with subtle differences
- 📈 Billions of phishing emails sent daily
- 🤖 Can we teach an SLM to detect phishing pages from URL and HTML?

1 The Dataset: PhreshPhish

📦 HuggingFace: phreshphish/phreshphish

🌐 Real-world phishing + benign webpages (500k samples)

📄 Each sample: URL + raw HTML + label (phish/benign)

Data Preparation Pipeline

Load & Sample

5,000 samples
from HF stream

→

Clean HTML

Strip scripts/styles
Keep key tags
~96% reduction

→

Length Filter

Max 5K chars
~4,800 samples

→

Language Filter

English only
~4,000 samples

→

Balance

Downsample
majority
~1,700 per class

2 Why Not Just Use GPT-5.4, Claude Opus 4.7, etc.?

SOTA APIs can solve this, but…

Sending raw HTML to third-party APIs leaks personal data

At scale: billions of pages/day = extreme latency & cost

Our choice: Qwen3-0.6B-4bit

A small, fine-tuned model runs fast, and privately

0.6B params, fits on a single GPU

* Same pipeline scales to larger models

GPT-5.4 Family on This Task

79.5%
gpt-5.4-nano

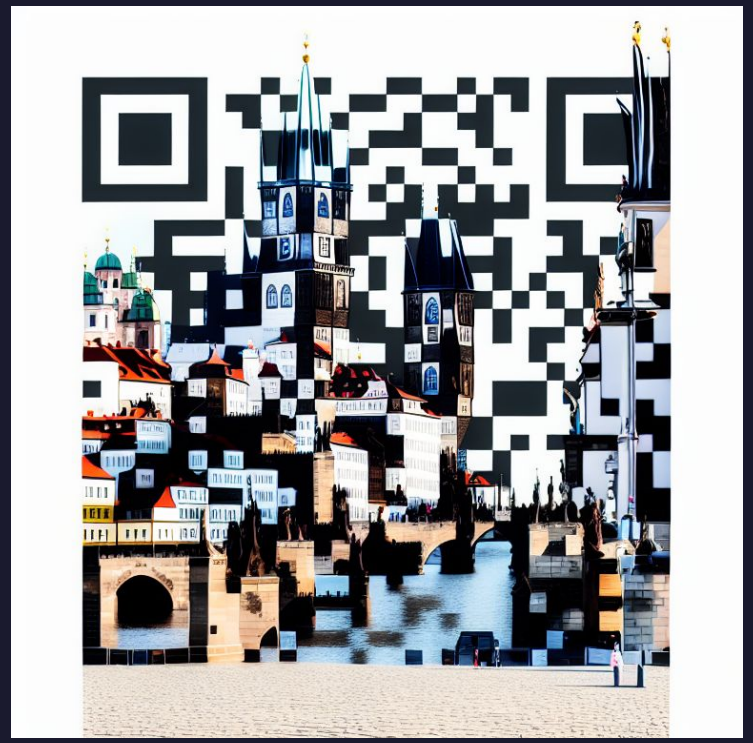
87.0%
gpt-5.4-mini

91.4%
gpt-5.4

Good — but not free, not private

2 Baseline Evaluation: Let's Run It!

How well does the base model detect phishing without any fine-tuning?



Environment setup:

- Scan the QR code (https://tinyurl.com/mlprague-dream)
- Click "**Copy to Drive**"
- (Make sure you're logged in to Gmail)

- Free-tier Google Colab notebook
- T4 GPU (released in 2018)

Baseline Results: The Model Fails

~58%	~85%	~11%	~20%
Accuracy	Precision	Recall	F1 Score

The model predicts "benign" for almost everything

Performance is weak, we need fine-tuning!

3

Hands-On: Exploration

Can you improve the base model's accuracy?

Prompting Techniques

Few-shot Prompting

Provide specific examples of the desired output pattern within the prompt to guide the model's response.

Example Prompt:

```
Classify the alert severity:
"Failed login..." -> Medium
"Employee clicked phishing..." -> High
"Port scan detected" -> Medium

Alert: Multiple failed admin login.
```

Response:

High

Chain-of-Thought (CoT)

Instruct the model to "think step-by-step" or reason through the logic before providing a final answer.

Example Prompt:

```
Classify the alert severity. Think step-by-step.
Alert: Multiple failed admin login.
```

Response:

1. Admin accounts are high-value targets.
2. Multiple failures suggest brute force.
3. Foreign IP adds significant suspicion.

Therefore, this alert should be classified as High severity.

High

Ideas to Explore

🎯 Add few-shot examples to the prompt

🧪 Test chain-of-thought prompting

🔍 Inspect phish vs. benign samples

✂️ Try different HTML cleaning strategies

📝 Experiment with different prompts

📏 Add a persona to the prompt

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Key insight: prompt engineering helps, but has limits.
The model doesn't truly understand the task yet.

4

SFT for Phishing Detection

Train the model on labeled examples of our task.

Input: prompt with URL + HTML.

The model learns to predict the expected output: phish or benign.

⚡ Let's Train!

Run the training cell and watch the loss go down

```
trainer.train()
```

5 Before vs. After Fine-Tuning

BEFORE (Base Model)		AFTER (Fine-Tuned)	
Accuracy	~58%	Accuracy	~93%
Precision	~85%	Precision	~96%
Recall	~11%	Recall	~89%
F1 Score	~20%	F1 Score	~92%

Key Takeaways

- SLMs are awesome!
- Tiny SLMs are often not task-ready out of the box.
- Prompt engineering helps, but has fundamental limits.
- SFT is surprisingly effective with small data + few steps.
- The full pipeline: data prep → prompt design → train → evaluate

Where to Go From Here

- Pick harder tasks & more diverse training data
- Scale to larger models
- Try different model families
- Experiment with different training methodologies (e.g. pre-training, full fine-tuning, Reinforcement Learning)

Thank You!

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Questions?

4

What is Supervised Fine-Tuning?

Human: Who was the first president of the United States?

Assistant: George Washington.

Human: When was he born and who was he married to?

Assistant: George Washington was born on February 22, 1732, and he was married to Martha Dandridge Custis.

4

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We train on assistant responses